

The definition of a quasi-category

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Contents

1	Introduction	2
2	Motivation	2
3	Simplicial Sets	3
4	Horns	3

1 Introduction

In this talk, we will introduce quasi-categories as models for ∞ -categories. This talk is technically accessible to the average undergraduate student with some basic upper-level training in mathematics, but more realistically the audience member should have had exposure to category theory, and if possible, some algebraic topology and simplicial homotopy.

A good reference for basic category theory is [ML98], and a good reference for the basic theory of simplicial sets is my own paper [Tal25].

2 Motivation

Consider a topological space X . We are often concerned not with studying X up to isomorphism, but rather up to weak homotopy equivalence, called the *homotopy type* of X . Recall that:

Definition 2.1. A *weak homotopy equivalence* is a continuous map $f : X \rightarrow Y$ between topological spaces that induces a bijection on path components $\pi_0(f) : \pi_0(X) \cong \pi_0(Y)$, and for each $i > 0$ and $x \in X$, induces an isomorphism of groups

$$\pi_i(f) : \pi_i(X, x) \cong \pi_i(Y, f(x)),$$

where we recall that

$$\pi_i(X, x) = \mathcal{T}\mathrm{op}_*((S^n, 0), (X, x)) / \sim.$$

A more succinct way of expressing the homotopical information of a topological space X is via its fundamental n -groupoids. Recall that a groupoid is a category in which all morphisms are invertible. Define $\Pi_1(X)$ to be the category where objects are given by points in X , and morphisms are given by homotopy classes of paths between these points.

The reason we cannot consider the paths themselves to be the morphisms is since path concatenation is *not associative*, but since it is associative up to homotopy, we get a category structure by considering homotopy classes of paths. However, we *want* to be able to consider the paths themselves, that way we get access to more information. For this to be possible, we have to work in a *weak 2-category*, which is a type of 2-category, meaning we not only have objects and (1-)morphisms, but we have *2-morphisms*, which are morphisms between our 1-morphisms. The relevance is that in a weak 2-category, compositions need not be well-defined (and to this end, need not be associative), but rather they only need to be well-defined up to 2-isomorphism.

The way this is relevant to our example is we can construct a 2-groupoid $\Pi_2(X)$, where 1-morphisms are paths, and 2-morphisms are homotopies. Since concatenation of paths is well-defined up to homotopy, and homotopies always admit inverses, this gives us our desired structure. *However*, this new structure gives us a new issue: just like concatenation of paths, composition of homotopies is *not associative*. It is only associative up to homotopy, often called a higher homotopy. We thus cannot consider actual homotopies as our 2-morphisms, but must instead consider homotopy *classes* as our 2-morphisms. In attempting to run away from our original problem, we have encountered it again, just in a new form.

Attempting to add higher homotopies will clearly yield the same issue. The fundamental n -groupoid $\Pi_n(X)$ has objects given by X , 1-morphisms given by paths, k -morphisms given by homotopies of $(k-1)$ -morphisms for $1 < k < n$, and n -morphisms given by homotopy classes of homotopies of $(n-1)$ -morphisms. We always run into the issue that the highest level has to be

homotopy *classes*, rather than just homotopies. It would be really nice if we could take some sort of limit as $n \rightarrow \infty$, obtaining the *fundamental* ∞ -groupoid $\Pi_\infty(X)$, which has n -morphisms for *all* $n \in \mathbb{N}$. Indeed, this is possible, and it is the goal of this talk to illustrate the construction of not just ∞ -groupoids, but $(\infty, 1)$ -categories, using the theory of quasi-categories.

Before we continue, some remarks on what to expect from this talk. Constructing categories which have n -morphisms for all n is extremely difficult, and is simplified greatly by instead constructing (∞, n) -categories, which have all k -morphisms, but for $k > n$, these morphisms are invertible. Categories of this form, especially $(\infty, 1)$ -categories, are extremely important in modern homotopy theory, and for this reason the term “ ∞ -category” has come to mean the same thing as “ $(\infty, 1)$ -category”. A category without the invertibility restriction is usually called an ω -category, since ω is the first infinite ordinal.

There are a few different constructions of ∞ -categories. We will use the theory of *quasi-categories*, which were originally introduced by Boardman and Vogt in [?], and were later pioneered by André Joyal in [Joy08a, Joy08b], and Lurie in [Lur09].

3 Simplicial Sets

Quasi-categories are built atop the theory of *simplicial sets*. We review the basic theory.

Definition 3.1. The simplex category Δ is defined as

- Objects are finite nonempty ordinals $[n] = \{0, \dots, n\}$;
- Morphisms are monotone functions, meaning $f : [n] \rightarrow [m]$ is a morphism if $x \leq y$ implies $f(x) \leq f(y)$.

4 Horns

Now that we understand some basic definitions, we would like to learn about *horns*. Horns are fundamental to the notion of a quasi-category, but they are nontrivial objects to understand. We begin with the notion of a simplicial subset.

Definition 4.1. Let X be a simplicial set. A *simplicial subset* of X is a simplicial set Y such that

- $Y_n \subseteq X_n$ for all n ;
- The faces and degeneracies of Y agree with those of X :

$$d_i^X|_{Y_n} = d_i^Y, \quad s_i^X|_{Y_n} = s_i^Y.$$

If Y is a simplicial subset of X , we write $Y \subseteq X$.

Note that definition 4.1 tells us what the faces and degeneracies of a simplicial subset must be. To construct a simplicial subset, we only need to specify the sets Y_n , and the subset condition gives us the faces and degeneracies for free.

Warning 4.2. Not all choices of subsets $Y_n \subseteq X_n$ yield a simplicial subset. For example, let $X_n = \mathbb{Z}$ for all n , and choose $Y_n = X_n$ for all $n \neq 1$, and $Y_1 = \emptyset$. There must be a degeneracy map $s_0 : Y_0 \rightarrow Y_1$, but there are no functions with an empty codomain. In particular, one must check the images of the faces and degeneracies lie within the sets Y_n .

Definition 4.3. Let $\alpha : X \rightarrow Y$ be a morphism of simplicial sets. Then there is a subsimplicial set $\text{im } \alpha \subseteq Y$, called the *image* of α , with sets given by

$$(\text{im } \alpha)_n = \text{im}(\alpha_n : X_n \rightarrow Y_n),$$

where we define α_n by recalling ??.

Many important simplicial sets arise as simplicial subsets of the standard n -simplicial set Δ^n . The first one is the *face*.

Definition 4.4. Note by the Yoneda lemma that postcomposition by a coface gives a morphism of simplicial sets $(d^i)_* : \Delta^{n-1} \rightarrow \Delta^n$. Define the i th face of Δ^n , denoted by $\partial_i \Delta^n$, as the image $\text{im}(d^i)^*$.

Remark 4.5. An important thing to note about the face is that since the epi-mono factorization in $\mathbf{\Delta}$ is unique, there is an isomorphism $\Delta^{n-1} \cong \partial_i \Delta^n$ given in one direction by postcomposing by d^i , and in the other by simply removing it. Under this isomorphism, the $(n-1)$ -simplex $1_{[n-1]} \in \Delta_{n-1}^{n-1}$ corresponds to the $(n-1)$ -simplex $d^i \in \partial_i \Delta^n$. By ?? and the Yoneda lemma, morphisms $\partial_i \Delta^n \rightarrow X$ are precisely morphisms $\Delta^{n-1} \rightarrow X$, which are precisely $(n-1)$ -simplices $\tau \in X_{n-1}$. Since morphisms $\Delta^{n-1} \rightarrow X$ are given by $1_{[n-1]} \mapsto \tau$, we have that morphisms $\partial_i \Delta^n$ are given by $d^i \mapsto \tau$.

Proposition 4.6. Let $\{X^i\}_{i \in I}$ be a collection of simplicial sets X^i such that their faces and degeneracies all agree with each other on levelwise intersections:

$$d_k^i | \bigcap_{i \in I} X_n^i = d_k^j | \bigcap_{i \in I} X_n^i$$

for all $i, j \in I$ and $d_k^i : X_n^i \rightarrow X_{n-1}^i$ the k th face. Then the levelwise union

$$\bigcup_{i \in I} X^i$$

is a simplicial set, and X^i is a simplicial subset for all i .

Proof. Indeed, by [ML98, §5.3 Thm. 1], limits in functor categories such as $\mathbf{sSet}$ are computed levelwise. Since the union of sets is the pushout along inclusions of the intersection into component sets

$$\begin{array}{ccc} X_n^i \cap X_n^j & \hookrightarrow & X_n^i \\ \downarrow & & \downarrow \\ X_n^j & \longrightarrow & X_n^i \cup X_n^j \end{array}$$

(and this generalizes for arbitrary unions), the statement is proven by cocompleteness of $\mathbf{Set}$. ■

The construction of the union of simplicial sets yields a simplicial set of fundamental importance: the *horn*.

Definition 4.7. Let $0 \leq i \leq n$. The i th horn Λ_i^n is defined as the simplicial subset

$$\Lambda_i^n := \bigcup_{j \in [n], j \neq i} \partial_j \Delta^n \subseteq \Delta^n.$$

Terminology 4.8. Let K be a simplicial set, and $\sigma_0 : \Lambda_i^n \rightarrow K$ a morphism of simplicial sets. We often refer to σ_0 as a *horn* in K . Similarly, we refer to a map $\partial_i \Delta^n \rightarrow K$ as a *face* of K .

For geometric intuition for horns, see [Tal25, §3.4]. We continue

References

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